

## LAB OUTLINE: PHYSICS I LAB (PHY108L)

### NORTH SOUTH UNIVERSITY

#### DEPARTMENT OF MATHEMATICS & PHYSICS

**Course** : Physics II Lab

**Course Code** : PHY 108L

**Credit** : One (01)

**Course Description** This lab course is designed to introduce variety of phenomenon in electromagnetism at the freshman level. The overall aims of these labs are to help the students to learn the basics of measuring different electrical and magnetic quantities and draw conclusion from the measured data.

#### Lab Outline

|              |                                                                      |
|--------------|----------------------------------------------------------------------|
| <b>Lab 1</b> | Verification of Ohm's Law                                            |
| <b>Lab 2</b> | Charging and discharging characteristics of a capacitor              |
| <b>Lab 3</b> | Identification of the pattern of magnetic fields due to a bar magnet |
| <b>Lab 4</b> | Current Behavior in an RL circuit                                    |
| <b>Lab 5</b> | Induced EMF and working principal of a transformer                   |
| <b>Lab 6</b> | Observation of electrical resonance in RLC circuit                   |

#### Mark Distributions

| COMPONENT                                                                     |                   | WEIGHT [%] |
|-------------------------------------------------------------------------------|-------------------|------------|
| Class Attendance                                                              |                   | 5%         |
| Class Work                                                                    | Class Performance | 5%         |
|                                                                               | Pre-Lab/Quiz      | 0 - 5%     |
|                                                                               | Lab Reports       | 55 - 60%   |
| Final Exam (this includes Set up, data collection & respond to questionnaire) |                   | 20%        |
| Viva                                                                          |                   | 10%        |